RV College of Engineering®
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Bengaluru - 560059, Karnataka, India

Jan-2025 - CIE - III

	DEPARTMENT OF MATHEMATICS		
Date	07.01.2025	**Time**	9.30 AM– 11.30 AM
Test	Improvement Test	**Maximum Marks**	10 + 50
Course Title	Statistics, Laplace Transform and Numerical Methods	**Course Code**	MA231TB
Semester	III	**Programs**	AS, BT, CH, ME, IEM

Sl. No	**Instructions:** Answer all questions.	M	CO	BT
	Part - A			
1	The real and imaginary parts of the complex valued function $f(z) = e^z$ is ________.	2	1	1
2	For the 1-D heat equation $\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2}$ subjected to the boundary conditions $u(0,t) = 0,\ u(4,t) = 0$ for $t > 0$ and initial conditions $u(x,0) = x(4-x)\ \ 0 < x < 4$, with $\alpha = 1/2$ and $h = 1$, then the value of $u_{1,2}$= ________.	2	3	2
3	Find the value of the constants a, b if the function $f(z) = (ax^4 + bx^2y^2 + y^4 + 2x^2 - 2y^2) + i(4x^3y - 4xy^3 + 4xy)$ is analytic.	2	2	2
4	For the one-dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, with the boundary conditions $u(0,t) = u(5,t) = 0$, and the initial conditions $u(x,0) = x(x-5)$ and $u_t(x,0) = 0$ with $h = 1$, $k = 1$, the value of $u(1,1)$ is ________.	2	3	2
5	The Taylor's series expansion of $f(z) = \cos z$ about $z = 0$ is ________.	2	2	1
	Part - B			
1	Determine the complex potential function $f(z) = u + iv$, given that $2u + v = e^x[\cos(y) - \sin(y)]$. Also find the velocity potential u and stream function v.	10	2	3
	Verify the function $v = x^2 - y^2 + \frac{x}{x^2+y^2}$ is harmonic.	3	1	2
b	Find an analytic function $f(z) = u + iv$ given that $v = \left(r - \frac{1}{r}\right)\sin\theta, r \neq 0$. Also find u.	7	2	2
3	Represent $\frac{1}{(z+1)(z+3)}$ as Laurent's series valid in the region (i) $1 < \|z\| < 3$ (ii) $\|z\| > 3$ (iii) $0 < \|z+1\| < 2$ (iv) $\|z\| < 1$ (ii)	10	4	2
4	The transverse displacement u of a point at a distance x from one end and at any time t of a vibrating string satisfies the equation $16\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial t^2} = 0$, with boundary conditions $u(0,t) = u(5,t) = 0$ and the initial conditions	10	3	3

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DEPARTMENT OF MATHEMATICS

	$u(x,0)=\begin{cases}3x, & 0\le x\le \frac{5}{2}\\ 20-4x, & \frac{5}{2}\le x\le 5\end{cases}$ and $\frac{\partial u}{\partial t}=0$ at $t = 0$. Solve the equation numerically up to $t = 1$, taking $h = 1$ and $k = 1/4$.		
5	Determine the values of $u(x,t)$ satisfying the parabolic equation $\frac{\partial u}{\partial t}=4\frac{\partial^2 u}{\partial x^2}$ and the boundary conditions $u(0,t)=u(8,t)=0$ and $u(x,0)=4x-\frac{x^2}{2}$ at the points $x=i: i=0,1,2,\ldots,8$ and $t=\frac{j}{8}: j=0,1,2,\ldots,5$.	10	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz & Test	Max Marks	5	21	24	10	04	36	20	-	-	

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date		**Maximum Marks**	**50**
Course Code	**IM233AI**	**Duration**	**90 Min**
Sem	**III Semester**	**Improvement Test**	

WORK SYSTEMS DESIGN

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Sketch a typical physical work system.	02	2	CO1
2	Distinguish between production and productivity.	02	2	CO1
3	Illustrate the pyramidal structure of a task.	02	2	CO1
4	What is meant by the term Poka-Yoke?	02	2	CO2
5	List the five steps in 5S system.	02	1	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1	A work group of 20 workers in a certain month produced 8600 units of output working 8 hr/day for 21 days. (i) What is the labor productivity ratio for this month? (ii) In the next month, the same work group produced 8000 units but there were 22 workdays in the month and the size of the work group was reduced to 14 workers. What is the labor productivity ratio for this second month? (iii) What is the productivity index using the first month as a base?	10	4	CO1
2	The standard time is to be established for a manual work cycle by direct time study. The observed time for the cycle averaged 4.80 minutes. The worker's performance was rated at 90% on all cycles observed. After eight cycles, the worker must exchange parts containers which took 1.60 minutes rated at 120%. The PFD allowance for this class of work is 15%. Determine the normal time, the standard time for the cycle and the worker's efficiency if the worker produces 123 work units during an 8 hour shift.	10	4	CO1
3	In Toyota Production System a standardized work procedure for a given task has three components. What are they? Explain any two in detail.	10	2	CO2
4	Sketch the structure of Lean Production System. Explain the features of different elements of the structure.	10	3	CO2
5	Distinguish between Push System of production control and Pull System of production control. Support your answer with an appropriate illustration.	10	3	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	**Particulars**		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	**Max Marks**	06	04	--	--	02	08	--	--	--	--
	Test	**Max Marks**	20	30	--	--	--	10	20	20	--	--

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USN										

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**8th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM234AI**	**Duration**	**120 Min**
Sem	**III Semester**	**Improvement CIE**	

MANUFACTURING PROCESSES

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	For drilling operation, the drill rotates with_____ pressure if job is held on earth.	1	L1	CO2
2	Point angle of drill is also known as __________	1	L1	CO3
3	In drilling, depth of cut is ____________ the drill diameter	1	L2	CO2
4	Cemented carbides tools are generally manufactured by ___________ technique.	1	L1	CO3
5	A tool had life of 10min with cutting speed 300m/min. Find the cutting speed for the same tool to have a life of 150min. Taken-0.2 in Taylor's tool life equation.	2	L2	CO2
6	State the difference between up and down milling.	2	L2	CO2
7	What are the advantages of Twist drill specify any two	2	L1	CO3
	Part –B			
1	List cutting tool materials? Describe in brief, stating its principal characteristics & application of any four cutting tool materials	10	L2	CO2
2	Sketch and explain the nomenclature of a twist drill.	10	L2	CO3
3	Write a short note on the following drilling operations i. Reaming ii. Counter sinking	10	L1	CO3
4	Draw a neat diagram of a Radial Drilling Machine. Name all the parts and explain the principle of operation	10	L2	CO2
5	Explain following milling operation with relevant sketches: i. Form milling ii. Gang milling	10	L1	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	--	06	04		05	05	--	--	--	--
	Test		--	20	30	10	20	30	--	--	--	--

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Academic year 2024 – 2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING& MANAGEMENT

Date	9th January 2025	**Maximum Marks**	10 + 50
Course Code	IM235AI	**Duration**	120 Min
Sem	III Semester	Improvement CIE	

DIGITAL METROLOGY

Part – A

Sl. No.	Questions	M	BT	CO
1.	What are the key components of a CMM system?	02	L1	CO4
2.	How does a computer-aided inspection system improve the accuracy of measurements in CMMs?	01	L2	CO4
3.	What is the principle of Interferometry used in optical flats for surface measurement?	02	L2	CO3
4.	Define the Ten-Point Height Average Value in surface finish analysis.	01	L1	CO3
5.	How do progressive and periodic errors impact the performance of threaded components, in terms of their functionality and precision	02	L2	CO3
6.	How can alignment errors be detected using an autocollimator?	02	L2	CO4

Part – B

Sl. No.	Questions	M	BT	CO
1.	Describe the principle of interference and its role in precision measurement. Explain the construction and operation of a Tool Maker's Microscope, highlighting its significance in precise dimensional analysis.	10	L3	CO3
2.	Define Pitch Circle diameter, addendum and dedendum and represent on a gear profile. How can a Vernier Gear Tooth Caliper be used to measure the thickness of a gear tooth? Explain the procedure briefly.	10	L3	CO3
3.	Discuss the Tomlinson Surface Meter, its construction, principles of operation, and applications in surface finish measurement. Explain how this specialized instrument contributes to accurate surface finish analysis in engineering and manufacturing processes, highlighting its advantages and limitations.	10	L3	CO3
4.	Describe the Michelson Interferometer and its working principles in detail, highlighting its key components and the interference phenomena it utilizes. Discuss the applications of Michelson Interferometers in various fields	10	L3	CO4
5.	Discuss the different types of CMMs and their applications in industrial inspection.	10	L2	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	--	--	05	05	--	03	07	--	--	--	--
	Test		--	--	30	20	--	--	10	40	--	--	--

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DEPARTMENT OF
BIOTECHNOLOGY

Academic year 2024-2025 (Odd Sem)

Date	8th JANUARY 2025	Maximum Marks	60
Course Code	BT232AT	Duration	120 min
Sem	III Semester	Improvement Test	
Bio Safety Standards and Ethics (Basket course)			

Answer QUIZ in sequence only in first two pages of answer booklets

Sl. o	Questions(quiz)	M	BT-L	CO
1.	What is food processing?	1	1	3
2	What is Innovative Packaging? Give examples	1	1	3
3	What are the advantages of preservation of food?	1	1	3
4	What are flavouring agents used in food? Give examples	1	1	4
5	List out the health policies in India	1	1	4
6	What is food safety?	1	1	4
7	What is ethics in food industry?	2	1	4
8	List out 4 methods used for food analysis	2	1	4

Sl. o	Questions (Test)	M	BT-L	CO
1.	Classify food preservatives, explain the strategies used for various foods preservation.	10	2	3
2	Illustrate and classify on various food packaging methods used in food industry.	10	3	4
3	"*Meat and meat products derived from livestock treated with antibiotics and hormones can cause adverse effects on consumer's health*" justify the statement	10	3	3
4	Elaborate on smart packaging methods with suitable examples. Add a note on packaging materials	10	4	4
5	Write explanatory notes on: Clinical Ethics and Research Ethics	5+ 5	2	3

BT-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	--	--	33	27	10	20	20	10	---	--

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Department of Mechanical Engineering

CIE - Improvement Test

Date	January 2025	Maximum Marks	50+10
Course Code	ME232TB	Duration	120 Min
Course Name	Material Science for Engineers	**USN:**	

Part - A (Quiz)

Q. No.	Questions	M	BT	CO
1.	Mention the main purpose of annealing heat treatment process of ferrous materials.	1	L1	3
2.	In ________ type of annealing, the lamellar pearlite is transformed into globular type structure.	1	L1	3
3.	In pack (solid) carburizing ________ is used as energizer.	1	L1	3
4.	Give two examples for 1D nanomaterial.	2	L1	3
5.	What happens to the elastic modulus property, when the material brought to nanoscale?	1	L1	4
6.	Name two top-down methods of nano-material synthesis.	2	L2	4
7.	In sol-gel process the formation of an oxide network takes place through ________ reactions of a molecular precursor in the liquid.	1	L1	4
8.	Mesoporous materials are a class of nanomaterials characterized by their highly ordered porous structure with pore sizes ranging from to nanometers.	1	L1	4

Part-B (Test)

Q. No.	Questions	M	BT	CO
1. a)	Describe the normalizing heat treatment process with the help of a neat sketch.	05	L2	3
b)	Discuss the flame hardening process with a neat sketch.	05		
2.	Illustrate the construction of TTT diagram for eutectoid steel of 0.8% carbon with neat sketches and label various features on the diagram. Also state the applications of the TTT diagram.	10	L3	3
3. a)	Enumerate the purposes of post-processing heat treatment methods of electronic devices.	05	L3	3
b)	Explain the rapid thermal processing with a neat sketch with its advantages.	05		
4.	"Identify whether the chemical vapour deposition (CVD) is top-down or bottom-up approach". Justify your answer and explain the working principle with the help of neat sketch. State is applications and advantages.	10	L3	4
5.	Discuss the working principle and various modes of atomic force microscope (AFM) with neat sketches.	10	L3	4

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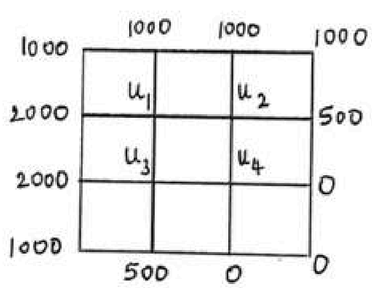
CIE - II

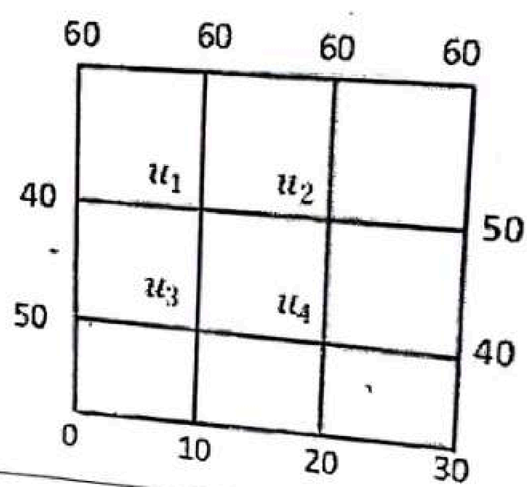
RV College of Engineering

DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester 2024)

Date	02.12.2024	Time	9.30 AM– 11.30 AM
TEST	Quiz- II & Test -II	Maximum Marks	10+50
Course Title	STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS	Course Code	MA231TB
Semester	III	Programs	AS, BT, CH, ME, IEM

Sl. No	Instructions: Answer all questions. Part - A	M	CO	BT
1	$L\left[\int_0^t \cosh t\right] =$ ________.	2	1	L1
2	If $f(t) = te^{-2t}$ then $L[f(t)\delta(t-4)] =$ __________.	1	1	L1
3	The inverse Laplace transform of $F(s) = \frac{s}{(s+a)^2}$ is ________.	2	1	L2
4	The convolution of functions $f(t) = e^t$ and $g(t) = t$ is ________.	2	2	L2
5	$L^{-1}\left[\frac{2}{(s-1)^2+4}\right] =$ ________.	1	1	L1
6	The initial approximation of u_3 in the Laplace equation $u_{xx} + u_{yy} = 0$ at the pivotal points of the given figure using standard five point formula and assuming $u_4 = 0$ is ________. 	2	1	L2
	Part - B			
1a	Determine the Laplace transform of the periodic function defined by $f(t) = \begin{cases} 1, & 0 \le t < a \\ -1, & a < t \le 2a \end{cases}$ and $f(2a+t) = f(a)$.	4	2	L2
1b	Express the following function in terms of unit step function and hence obtain its Laplace transform $f(t) = \begin{cases} t^2, & 1 < t \le 2 \\ 4t, & t > 2 \end{cases}$.	6	3	L3
2	Obtain the inverse Laplace transform of the function $F(s) = \frac{s}{(s+1)(s^2+1)}$ using the method of partial fractions.	10	2	L2
3	Employing convolution theorem, determine the inverse Laplace transform of $F(s) = \frac{s^2}{s^4-16}$.	10	3	L3

4	The displacement $x(t)$ of a mass-spring system is governed by $\frac{d^2x}{dt^2}+\lambda\frac{dx}{dt}+k\,x=f(t)$, where λ and k are force constants subject to the conditio $x(0)=2, \frac{dx}{dt}(0)=-4$ and $\lambda=2, k=5, f(t)=\frac{17}{2}\sin 2t$. Determine $x(t)$ by using Lapla transform method.
5	Given the values of $u(x,y)$ on the boundary of the square region. Evaluate t function $u(x,y)$ satisfying the Laplace equation $u_{xx}+u_{yy}=0$ at the pivotal poin of the figure. Perform four iterations. 

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BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L
	Quiz Max Marks	8	2			4	6			
	Test Max Marks	0	14	26	10	0	14	26	10	

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Academic year 2023-2024 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**20th March 2024**	**Maximum Marks**	**50**
Course Code	**IM233AI**	**Duration**	**90 Min**
Sem	**III Semester**	**Improvement Test**	
WORK SYSTEMS DESIGN			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	A work group of 12 workers in a certain month produced 1400 units of output working 8 hr/day for 18 days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same work group produced 1690 units but there were only 12 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	10	3	CO1
2	A worker performs a repetitive assembly task at a work branch at a work bench to assemble products. Each product consists of 25 components. Various hand tools are used in the task. The standard time for the work cycle is 7.45 minutes, based on using a PFD allowance factor of 15%. If the worker completes 75 product units during an 8 hour shift, determine the number of standard hours accomplished and the worker efficiency. If the worker took only one rest break, lasting 13 minutes and experienced no other interruptions during the 8 hours of shift time, determine the worker performance.	10	4	CO3
3	Illustrate the structure of Lean Production System. Explain the features of different elements of the structure.	10	3	CO2
4	List all the forms of waste as advocated by Taiichi Ohno in a production environment. Elaborate on any three forms in detail.	10	3	CO2
5	Summarize the prerequisites for a successful implementation of Just in Time philosophy.	10	2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	30	10	--	--	10	30	10	--	--

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Academic Year 2023-2024 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**20th March 2024**	**Maximum Marks**	**50**
Course Code	**IM234AI**	**Duration**	**90 Min**
Sem	**III**	**Improvement Test**	
MANUFACTURING PROCESS			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	Examine the geometry of a twist drill and its impact on drilling operations.	10	1	1
2	List three common types of drills used in a drilling machine. Provide a brief description of each drill type and their typical applications.	10	2	2
3	Differentiate between up milling and down milling.	10	3	3
4	Show the calculations for setting dividing head to mill 69 divisions on a spur wheel blank by compound indexing?	10	2	2
5	Determine time taken for plain milling a rectangular surface of length 100 mm and width 50 mm by a helical fluted plain HSS milling cutter of diameter 60 mm, length 75 mm and 6 teeth. Assume A = O = 5 mm, V_C = 40 m/min and so = 0.1 mm/tooth	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	**Particulars**	**CO1**	**CO2**	**CO3**	**CO4**	**L1**	**L2**	**L3**	**L4**	**L5**	**L6**
	Max Marks	**10**	**20**	**20**	--	**10**	**20**	**20**	--	--	--

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Academic year 2023-2024 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING& MANAGEMENT

Date	**22nd March 2024**	**Maximum Marks**	**50**
Course Code	**22IM325AI**	**Duration**	**90Min**
Sem	**III Semester**	**Improvement Test**	
DIGITAL METROLOGY			

Sl. No.	Questions	M	BT	CO
Q1	Explain the functioning of a tactile sensors and Piezoelectric sensor, detailing its basic operation and how it detects contact with surfaces. Provide examples of applications where these sensors are commonly used.	10	L3	CO2
Q2	Discuss six key specifications of sensors and describe briefly the operational principles of Potentiometer Sensors. Provide a detailed explanation of how Potentiometer Sensors function and illustrate their applications.	10	L3	CO2
Q3	Explain briefly the 4 main elements of a Coordinate Measuring Machine (CMM)	10	L2	CO4
Q4	How does Coordinate Measuring Machine (CMM) integration with Computer-Aided Manufacturing (CAM) enhance quality control processes in manufacturing? Discuss with examples.	10	L3	CO4
Q5	Describe the Michelson Interferometer and its working principles in detail, highlighting its key components and the interference phenomena it utilizes. Discuss the applications of Michelson Interferometers in various fields	10	L2	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	**Particulars**		**CO1**	**CO2**	**CO3**	**CO4**	**CO5**	**L1**	**L2**	**L3**	**L4**	**L5**	**L6**
	Test	**Max Marks**	--	20	--	30	--	--	20	30	--	--	--

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Department of Civil Engineering

Date	19-03-2024	Maximum Marks	50
Course Code	CV232AT	Duration	90 Min
Sem	III	CIE – III (Test)	
Environment and Sustainability			

Sl. No.		Questions	M	BT	CO
1	a	Enumerate the impacts of ozone layer depletion.	5	2	3
	b	Explain the concept of 5R.	5	2	3
2		Define the following. i) EIA ii) Climate change iii) Carbon Credit iv) Carbon Footprint v) Green Building	10	2	3
3		Explain the principles of CSR.	10	2	4
4		Enumerate the ways to reduce carbon foot print.	10	3	3
5	a	Briefly explain the concept of circular economy.	5	2	3
	b	Explain the concept of charity and corporate philanthropy.	5	2	4

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	-	35	15	-	40	10	-	-	-

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Department of Mechanical Engineering

CIE – Improvement

Date	March 2024	Maximum Marks	50
Course Code	ME232AT	Duration	90 Min
Course Name	Materials Science for Engineers	USN:	Sem: 3

Q. No.	Questions	M	BT	CO
1a.	Define heat treatment and explain the purpose of heat treatment.	5	2	3
1b.	Discuss the advantages and applications of rapid thermal processing for electronic devices.	5	2	3
2a.	Differentiate between annealing and normalising heat treatment processes.	5	3	3
2b.	Describe the full annealing heat treatment process for ferrous materials.	5	3	3
3a.	Explain the steps involved in thermal oxidation of semiconductor devices,	6	3	3
3b.	Describe the defects of heat treatment process for different materials.	4	2	3
4.	With the help of TTT diagram explain hardening process.	10	3	3
5.	Explain the following heat treatment processes with neat sketches: i) Flame hardening and ii) carburising	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	TEST Marks	-	-	50	-	-	14	36			

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DEPARTMENT OF BIOTECHNOLOGY

Academic year 2023-2024 (Odd Sem)

Sem	III Semester	Closed Book Offline Test-3
	Bio Safety Standards and Ethics (Basket course)	

Sl. No	Questions (Test)	M	BT-L	CO
1.	Explain on various food processing techniques	10	2	4
2	Illustrate on the following: Blanching Pasteurization Frying	10	4	4
3	Describe three levels of packaging the food products	10	4	4
4	Explain the effects of poor packaging on food products, add a note on smart packing	10	2	4
5	Write Explanatory notes on Freezing De hydration Homogenization	10	2	4

BT-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

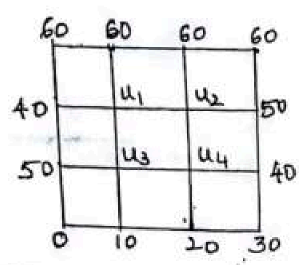
Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	--	--	---	50	---	30	--	20	---	--

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Mathematics

Course: Statistics, Laplace Transforms and Numerical Methods	**Improvement Test**	**Maximum marks: 50**
Course code: MA231TB	**III Semester (AS, BT, CH, IM, ME)**	**Time: 10.00AM-11.30AM** **Date: 19/03/2024**

Instructions to candidates: ***Answer all questions***

Q.No	Questions	M	BT	CO
1(a)	Find the Laplace transform of $e^{2t}(3t^5 - \cos 4t)$.	5	1	1
1(b)	Find the Laplace transform of $\frac{e^{at} - \cos bt}{t}$.	5	2	2
2(a)	Express the following function in terms of unit step function and hence find the Laplace transform of $f(t) = \begin{cases} 2t & for\ 0 < t < \pi \\ 1 & for\ \ \ t > \pi \end{cases}$.	5	2	2
2(b)	Solve the parabolic equation $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ with the conditions $u(0,t) = u(1,t) = 0,\ u(x,0) = x(1-x)$. Assume $\Delta x = h = 0.2$. Tabulate $u(x,t)$ for 3 levels of time using Bendre Schmidt Method.	5	3	3
3	Find the solution of one-dimensional wave equation $u_{tt} = u_{xx}$ up to $t = 0.5$ with a spacing of $\Delta x = h = 0.1$ and $\Delta t = k = 0.1$ subject to $u(0,t) = 0, u(1,t) = 0,\ u_t(x,0) = 0$ and $u(x,0) = 10 + x(1-x)$.	10	3	3
4	The temperature distribution $u(x,t)$ is modelled as one-dimensional heat equation $\frac{1}{4} u_t = u_{xx}$ for $0 \le x \le 4$ subject to the boundary conditions $u(0,t) = u(4,t) = 0$ and initial condition $u(x,0) = \frac{x(4-x)}{2}$. Determine the temperature distribution $u(x,t)$ by taking $\Delta x = h = 1$ and $\Delta t = k = \frac{1}{16}$ using Schmidt relation for 4 levels.	10	3	4
5	Solve the Laplace equation $u_{xx} + u_{yy} = 0$ given the square mesh (Carry out five iterations). 	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	5	10	25	10	5	10	35	--	-	-

***********ALL THE BEST***********

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Academic year 2022-2023 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**21st March 2023**	**Maximum Marks**	**10 + 50**
Course Code	**21IM33**	**Duration**	**20 + 90 Min**
Sem	**III Semester**	**CIE - Improvement**	

WORK SYSTEMS DESIGN

Note:
1. Answer all the Questions.

PART - A

Sl. No.	Questions	M	BT	CO
1	Standard Time is also known as Allowed Time. Justify.	02	2	3
2	Distinguish between First generation and Higher generation levels of PMTS.	02	2	3
3	List the prerequisites for valid time standards.	02	1	3
4	What are the broad two components of Environment in the context and scope of Ergonomics analysis?	02	1	4
5	Classify the three basic categories of Human-Machine systems.	02	2	4

PART - B

Sl. No.	Questions	M	BT	CO
1.	Review different types of Allowances used in Time Standards. Justify the need to consider the same in determining standard times.	10	4	3
2.	The observed average time in a direct time study was 2.40 minutes for a repetitive work cycle. The worker's performance was rated at 110% on all cycles. The personal time, fatigue and delay allowance for this work is 12%.Determine the normal time and the standard time for the cycle.	10	3	3
3.	The snap back timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element *c* is a machine controlled and the time is constant. Elements *a,b,d,e* and *f* are operator controlled and external to machine cycle and they are performance rated at 80%.If the machine allowance is 25% and worker PFD allowance is 15%, determine the normal time and standard time for the work cycle. Element: a 0.18, b 0.30, c 0.88, d 1.12, e 1.55, f 1.80 (Observed Time (min))	10	4	3
4.	Illustrate and Explain through a block diagram model, the typical interactions in a human-machine system.	10	3	4
5.	Distinguish between the terms, Hazard and Accident. Also illustrate a model depicting the possible factors that contribute to industrial accidents.	10	3	4

Element	a	b	c	d	e	f
Observed Time (min)	0.18	0.30	0.88	1.12	1.55	1.80

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	--	--	30	20	--	--	--	30	20	--	--
	Quiz		--	--	06	04	--	04	06	--	--	--	--

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Date	21st March 2023	Maximum Marks	10 + 50
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Sem	III Semester	CIE - Improvement	

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PART - B

Sl. No.	Questions	M	BT	CO
1.	Review different types of Allowances used in Time Standards. Justify the need to consider the same in determining standard times.	10	4	3
2.	The observed average time in a direct time study was 2.40 minutes for a repetitive work cycle. The worker's performance was rated at 110% on all cycles. The personal time, fatigue and delay allowance for this work is 12%.Determine the normal time and the standard time for the cycle.	10	3	3
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Table for Question 3:

Element	a	b	c	d	e	f
Observed Time (min)	0.18	0.30	0.88	1.12	1.55	1.80

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	--	--	30	20	--	--	--	30	20	--	--
	Quiz		--	--	06	04	--	04	06	--	--	--	--

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Academic year 2022-2023 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	21st March 2023	Maximum Marks	10+50
Course Code	21IM34	Duration	20+90 Min
Sem	III Semester	Improvement Test	

MANUFACTURING PROCESS

Note:
1. Answer all the Questions.

PART A

QNo.1	Questions	M	BT	CO
1.1	In chemical machining is material removal happens with the help of _______	01	L2	CO3
1.2	In the Electro Chemical Machining process, there is the tool and work piece. Which is the Cathode in the configuration?	01	L1	CO4
1.3	List any two features of advanced machining process	02	L2	CO3
1.4	In AJM, which of the materials are used as abrasive grains?	02	L1	CO2
1.5	Define Hybrid Machining processes	02	L2	CO3
1.6	Summarize difference between EBM and LBM	02	L3	CO4

PART B

Sl. No.	Questions	M	BT	CO
1.	Classify the different types of advanced Machining Processes used in the Industries	10	L2	CO3
2.	Draw the schematic diagram and explain the principle of operation of electrochemical grinding process.	10	L2	CO2
3.	With the aid of sketch, explain the working principle of Water-jet Machining.	10	L3	CO4
4.	Describe the working principle and construction of laser jet machining.	10	L2	CO3
5.	State the differences between additive and subtractive manufacturing	10	L3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	-	02	05	03	02	06	02	-	-	-
	Test		-	10	20	20	-	30	20	-	-	-

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DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	21st March 2023	Maximum Marks	10+50
Course Code	21IM34	Duration	20+90 Min
Sem	III Semester	Improvement Test	
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Note:

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BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	-	02	05	03	02	06	02	-	-	-
	Test		-	10	20	20	-	30	20	-	-	-